Step-Signal-Injection-Based Robust MTPA Operation Strategy for Interior Permanent Magnet Synchronous Machines

Jinhui Xia, *Student Member, IEEE*, Yuanbo Guo, *Member, IEEE*, Ze Li, *Student Member, IEEE*, Juri Jatskevich, *Fellow, IEEE*, and Xiaohua Zhang

***Abstract*—In various applications that utilize maximum torque per ampere (MTPA) operation of interior permanent magnet synchronous machines (IPMSMs), there exist unexpected perturbations in electrical parameters and operating environment, which deteriorate the accuracy and efficiency of the MTPA operation. This paper establishes a step-signal-injection-based robust MTPA operation strategy for the IPMSM drives. The method works by injecting a step signal into the current vector angle and observing the response in the current magnitude, followed by a proportional-integral controller which returns the system to optimal operation. The stability of the proposed algorithm is established using the Lyapunov theory. A speed-servo control system of IPMSM is considered, where a disturbance detection unit is designed to switch operation between optimal current angle updating mode and steady-state MTPA operation mode. Since the optimal current angle updating is independent of machine parameters, the impact of system perturbations on the MTPA operation can be effectively suppressed. Extensive experimental results for IPMSM and hardware-in-the-loop-based machine are presented to validate the effectiveness and robustness of the proposed method.**

***Index Terms*—Interior permanent magnet synchronous machine, MTPA operation, optimal current vector angle, step signal injection.**

I. Introduction

INTERIOR permanent magnet synchronous machines (IPMSMs) are increasingly considered in emerging electric traction and vehicular propulsion applications due to their high efficiency, high torque density, and high reliability characteristics [1]. To obtain a better control performance, various control methodologies have been proposed in the literature, among which the maximum torque per ampere (MTPA)-based methods are often considered with IPMSMs. Through fully utilizing the reluctance torque caused by the rotor saliency in IPMSMs, higher efficiency and operating stability can be achieved under MTPA operation [2].

Generally, optimal current vector angle calculation is required for MTPA operation of the IPMSM, wherein several electrical parameters including stator inductances and permanent magnet flux linkage are needed to determine the optimal current angle. Due to temperature variations, magnetic saturations, cross-coupling effects, and identification errors, these parameters may vary under different operating conditions, resulting in undesired deviation of the current angle from its optimal value and degradation of the MTPA operation [3]–[5].

To mitigate these problems, numerous online MTPA tracking methods have been proposed, which can be categorized as Look-up table (LUT)-based [2], [6]–[10]; system-reconfiguration-based [11]–[16]; and signal-injection-based methods [17]–[23], [25]–[30], respectively. In the LUT-based methods, the formula of the optimal current angle is simplified and represented as stored LUT data (or approximated as a function of current and operating temperature). However, the results derived from the LUT are still approximate, and the forming of accurate LUT for wide range of operating conditions may be challenging. The control for IPMSM can be reconfigured to directly generate the optimal voltage reference combinations as in [13]–[16]. However, useful physical parameters such as speed and currents are not always available in such methods. The real time tracking and calibration of these physical parameters are necessary for many applications with electric transportation and propulsion. Hence, system-reconfiguration-based methods may not be suitable for such applications.

Due to many practical challenges with the previous two methods, the signal-injection-based approaches are becoming attractive alternatives. For example, in virtual-signal-injection-based methods [17]–[23], the stability of the control will not be affected by the injection, since no actual signals are being injected. Nevertheless, the information about the machine's

Manuscript received February 10, 2019; revised June 11, 2019; accepted July 19, 2019. Date of publication July 31, 2019; date of current version November 21, 2019. This work was supported in part by National Natural Science Foundation of China under Grants 51407023 and 51377013 and in part by the Fundamental Research Funds for the Central Universities under Grant DUT15RC(4)04. Paper no. TEC-00150-2019. *(Corresponding author: Xiaohua Zhang.)*

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Digital Object Identifier 10.1109/TEC.2019.2932455

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stator resistance and electromagnetic torque is needed [17]–[22]. Therefore, estimation of parameters and operating conditions is challenging in these methods [24].

In direct-signal-injection-based methods, the MTPA operation is realized by injecting the sinusoidal or square-wave signals into the current vector angle, the current references, or the voltage space vectors, and observing/measuring and processing their responses [25]–[30], which has some practical advantages. Specifically, many parameters of IPMSM are not required, which enhances the robustness of such methods under parameter uncertainties. An automatic MTPA tracking method has been proposed in [25], where a sinusoidal signal is injected into the current vector angle, and the MTPA operation is achieved using a closed-loop control. In methods that operate by performing online search for the minimum current point in a closed-loop, various additional digital filters are commonly employed in control loops in order to extract useful information after the signal injection [27]–[30]. However, the bandwidth of controllers may become a limiting factor.

A review of state-of-the-art existing methods shows that most of them are highly dependent on machine's parameters. Moreover, complicated digital filters and data processing algorithms are not desirable due to their impact on the controller bandwidth, real-time performance, and cost of the overall drive system. To overcome these drawbacks, in this paper, a robust MTPA control strategy for IPMSM is proposed. The main idea of the proposed algorithm is to perturb the angle of the stator current vector using a step signal with adequate amplitude and sign, and then to measure the difference (which approximates derivative) of the stator current amplitude in two consecutive samples and regulate the current vector angle with a proportional-integral (PI) controller. The main contributions of this paper are summarized as follows:

1) A new step-signal-injection-based robust MTPA operation scheme is presented. The algorithm needs only the current magnitude (which can be directly obtained in the control system) for the optimal current angle calculation, which makes the algorithm independent of electrical parameters of the IPMSM.
2) A speed-servo control system of the IPMSM including an optimal current angle updating mode and a steady-state MTPA operation mode is considered, where a disturbance detection unit (DDU) is designed to determine the exact mode in which the system should operate.
3) The effectiveness and robustness of the proposed algorithm are evaluated experimentally using IPMSM and hardware-in-the-loop (HIL)-based platform. The results demonstrate that the proposed algorithm accurately follows the MTPA operation of the IPMSM under multiple system perturbations.

II. MTPA Operation for IPMSMs

A. Modeling of IPMSM

A three-phase IPMSM with the simplified cross-section diagram shown in Fig. 1 is considered in this paper. The machine is assumed to have negligible hysteresis and eddy-current losses.

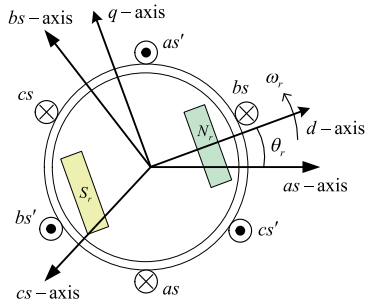


Fig. 1. Simplified cross-section diagram of the IPMSM.

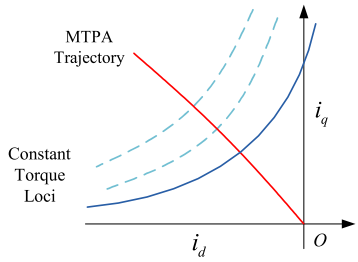


Fig. 2. Illustration of MTPA trajectory of an IPMSM in $i_{q,d}$ coordinates.

The voltage equations $v_{q,d}$ and the electromagnetic torque T_e can be expressed in *qd* rotor reference frame as [17]

$$\begin{cases} v_q = Ri_q + L_q \frac{d}{dt} i_q + \omega_e \left(L_d i_d + \lambda_f\right) \\ v_d = Ri_d + L_d \frac{d}{dt} i_d - \omega_e L_q i_q \\ T_e = \frac{3}{2} P i_q \left[\left(L_d - L_q\right) i_d + \lambda_f\right] \end{cases}, \tag{1}$$

where $i_{q,d}$ are the stator currents; ω_e is the electrical angular velocity of the machine; λ_f is the flux linkage of the permanent magnet; $L_{q,d}$ are the stator inductances in *qd* reference frame; and *P* is the number of pole pairs.

B. Conventional MTPA Control Strategy

As it is often the case for IPMSM, the *d*-axis contains the permanent magnets (as illustrated in Fig. 1), which reduce the permeability in *d*-axis direction and result in a magnetic saliency (i.e., $L_d < L_q$). Therefore, as it can be observed in (1), there exist two components contributing to the overall electromagnetic torque T_e, which are the field torque resulting from the interaction of the permanent magnet flux linkage λ_f with current i_q, and the reluctance torque produced by the saliency of the IPMSM.

Theoretically, for a given torque T_e, there exist an infinite number of $i_{q,d}$ combinations that satisfy (1). The conventional field-oriented control method sets $i_d = 0$, which eliminates the reluctance torque, and therefore is not suited for PMSMs with salient poles. Hence, the MTPA strategy minimizes the stator current magnitude by making use of both the field torque and the reluctance torque. The MTPA operation of an IPMSM can be illustrated in Fig. 2, which shows the curves in $i_{q,d}$ coordinates corresponding to certain constant torque values.

The MTPA trajectory corresponds to the minimum stator current magnitude for any given torque level. This optimal trajectory can be derived by searching for the minimum distance

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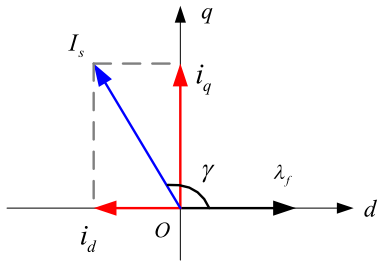


Fig. 3. Diagram of the current vector in *qd* coordinates.

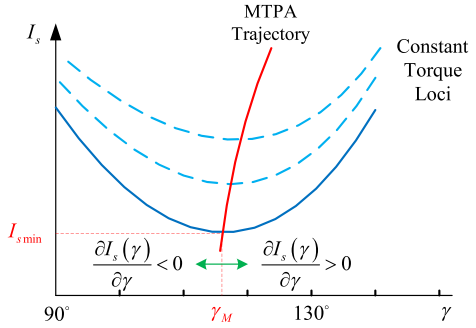


Fig. 4. MTPA trajectory in $I_s - \gamma$ coordinates.

from the torque locus to the origin of the coordinate depicted in Fig. 2. Using the MTPA control, the efficiency under a given load can be increased by minimizing the copper and conduction losses.

The diagram of the stator current vector in *qd* reference frame is depicted in Fig. 3. The relationship between *q*- and *d*-axes current projections and the magnitude I_s can be obtained as

$$i_q = I_s \sin\gamma,\ i_d = I_s \cos\gamma, \tag{2}$$

where γ is referred to as the current vector angle. After finding the minimum current magnitude, the projections i_q and i_d and the corresponding optimal current vector angle γ can be obtained based on Fig. 3. The optimal current angle γ_M that generates the minimum stator current magnitude under a given torque can be calculated as [13]

$$\gamma_M = \arccos\left(\frac{-\lambda_f + \sqrt{\lambda_f^2 + 8(L_d - L_q)^2 I_s^2}}{4\left(L_d - L_q\right) I_s}\right). \tag{3}$$

However, in practical applications, the inductances $L_{q,d}$ and the flux linkage λ_f required in (3) may deviate due to operating temperature variations and magnetic saturation. In such conditions, the calculated current angle γ_M and the corresponding combination of $i_{q,d}$ will drift away from the MTPA optimal point, resulting in a decrease in the operating efficiency. This issue leads to the need of a robust MTPA control method that is independent of machine parameters.

III. Proposed Step-Signal-Injection MTPA Control

A. Algorithm Design

Substituting (2) into (1), the torque is rewritten as

$$\begin{aligned} T_e &= \frac{3}{2} P \lambda_f I_s \cos\gamma + \frac{3}{4} P \left(L_d - L_q\right) I_s^2 \sin 2\gamma \\ &= k_1 I_s \cos\gamma + k_2 I_s^2 \sin 2\gamma, \end{aligned} \tag{4}$$

where k_1 and k_2 are two coefficients defined as

$$k_1 = \frac{3}{2} P \lambda_f,\ k_2 = \frac{3}{4} P \left(L_d - L_q\right). \tag{5}$$

Based on (4), the stator current magnitude as a function of stator current vector angle can be obtained as

$$I_s(\gamma) = \frac{-k_1 + \sqrt{k_1^2 + 8 k_2 T_e \tan\gamma}}{4 k_2 \sin\gamma}. \tag{6}$$

Function $I_s(\gamma)$ is convex with only one minimum extrema point for the angle γ between 90° and 180°. The MTPA trajectory in $I_s - \gamma$ coordinates is demonstrated in Fig. 4, where the relationship between the magnitude I_s and the angle γ can be observed from the constant torque locus.

As it can be seen in Fig. 4, before the current magnitude reaches the lowest point $I_{s\min}$ from the left ($\gamma < \gamma_M$), the value of $\partial I_s(\gamma)/\partial\gamma$ is negative. With the increasing of current angle ($\gamma > \gamma_M$), the value of $\partial I_s(\gamma)/\partial\gamma$ becomes positive. For steady-state operation under a given load torque T_L, the online calculation of the optimal current angle γ_M becomes a constrained optimization described as

$$\min_{\gamma} |I_s(\gamma)| \text{ subject to } T_e\left(I_s(\gamma), \gamma\right) = T_L. \tag{7}$$

If the IPMSM parameters are unknown or deviated from their nominal values, the current angle calculated using (3) (with incorrect parameters) will not be optimal.

In the proposed method, a step signal γ_h with the amplitude of *A* is injected into the current vector angle γ. After the injection, the response $I_s(\gamma + \gamma_h)$ is observed, and it can be expressed using Taylor series as

$$\begin{aligned} I_s(\gamma + \gamma_h) &= I_s(\gamma) + A\frac{\partial I_s(\gamma)}{\partial \gamma} + \frac{1}{2}A^2\frac{\partial^2 I_s(\gamma)}{\partial \gamma^2} + \cdots \\ &\approx I_s(\gamma) + A\frac{\partial I_s(\gamma)}{\partial \gamma}. \end{aligned} \tag{8}$$

To achieve the convergence of the current vector angle to the actual optimal angle γ_M (see Fig. 4), a positive parameter *K* is introduced to ensure that the time derivative of γ satisfies [29]

$$\frac{d\gamma}{dt} = -K\frac{\partial I_s(\gamma)}{\partial \gamma}\ (K > 0). \tag{9}$$

As observed in (9), when $\partial I_s(\gamma)/\partial\gamma > 0$, the current angle γ will decrease with a negative time derivative. On the contrary, when $\partial I_s(\gamma)/\partial\gamma < 0$, the current angle γ will increase with a positive time derivative. The integral of (9) gives the equation for optimal current angle tracking by

$$\gamma = K_i A \int \frac{\partial I_s(\gamma)}{\partial \gamma} dt, \tag{10}$$

where K_i is an integral coefficient that satisfies $K_i = -K/A$. As it can be observed from Fig. 4, if the initial position of γ lies on the left of γ_M, a positive step $\gamma_h = A > 0$ should be injected for

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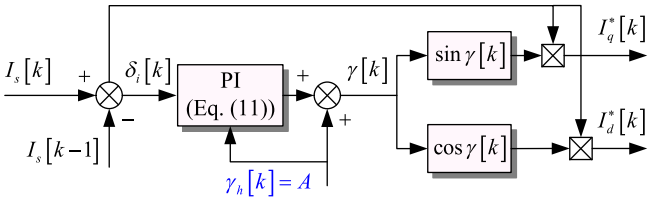


Fig. 5. Block diagram of optimal current angle tracking.

the optimal current angle to converge, and the integral coefficient K_i needs to be negative in terms of (9). Similarly, if γ lies on the right of γ_M initially, a negative step should be considered with a positive K_i. Therefore, the sign of K_i should be designed opposite to the sign of A.

Based on this feature, the block diagram of the optimal current angle tracking is designed and illustrated as Fig. 5, where the integral term has been improved as a PI controller. By substituting (8) into (10), the calculation of the optimal current angle can be derived and discretized as

$$\gamma[k] = \gamma_h[k] + \gamma[k-1] + (K_p + K_i T_s)\,\delta_i[k] - K_p \delta_i[k-1], \tag{11}$$

where K_p is the proportional gain of the controller, δ_i is the difference (error) of the current magnitude I_s between two adjacent sampling periods, and T_s is the sampling period.

As seen in Fig. 5, after the step signal $\gamma_h[k-1] = A$ is injected into the current vector angle $\gamma[k-1]$, the current magnitude response $I_s[k]$ can be observed in the kth sampling period. The PI controller will regulate the magnitude difference $\delta_i[k]$ and provide the current vector angle reference $\gamma[k]$ for generating the q- and d-axes current references of the IPMSM. It is noted that no parameters of the IPMSM are required in the optimal current angle tracking.

B. Stability Analysis and Parameter Selection

To verify the stability of the proposed algorithm, a Lyapunov function candidate V is designed as [25]

$$V = \frac{1}{2}\delta_i^2. \tag{12}$$

Since $I_s(\gamma)$ is a convex function, the second derivative of $I_s(\gamma)$ satisfies

$$\frac{\partial^2 I_s(\gamma)}{\partial \gamma^2} \approx \frac{1}{A}\frac{\partial\left(I_s(\gamma+\gamma_h) - I_s(\gamma)\right)}{\partial \gamma} = \frac{1}{A}\frac{d\delta_i}{d\gamma} > 0 \tag{13}$$

Based on (8), (10), and (13), the time derivative of V yields

$$\frac{dV}{dt} = K_i A \frac{\partial^2 I_s(\gamma)}{\partial \gamma^2}\delta_i^2 < 0. \tag{14}$$

Therefore, the stability of the proposed optimal current angle tracking algorithm can be proved.

To calculate the convergence time of the algorithm, differentiating δ_i with respect to time yields

$$\frac{d\delta_i}{dt} = \frac{d\delta_i}{d\gamma}\frac{d\gamma}{dt} = K_i A \frac{\partial^2 I_s(\gamma)}{\partial \gamma^2}\delta_i. \tag{15}$$

The differential equation (15) has the following basic solution

$$\delta_i(t) = \delta_0 e^{K_i K_{Is} A t}, \tag{16}$$

where δ_0 is the initial value of $\delta_i(t)$, and K_{Is} is a positive variable that satisfies $K_{Is} = \partial^2 I_s(\gamma)/\partial\gamma^2$. As seen in (16), $\delta_i(t)$ will gradually go to 0 with $K_i K_{Is} A < 0$. When the value of $\delta_i(t)$ drops within a small margin of error $\pm|\delta_M|$ during the convergence, the convergence time τ of $\delta_i(t)$ can be calculated by

$$\tau = -\frac{1}{K_i K_{Is} A}\ln\frac{\delta_M}{\delta_0}, \tag{17}$$

where δ_M is with the same sign as δ_0 in terms of (16). When selecting the magnitude A of the step, it is essential to ensure that the response of the injection will not be overwhelmed by the noises in the system. In order to select an appropriate value for K_i, (10) can be rewritten as

$$\frac{d\gamma}{dt} = K_i A \frac{\partial I_s(\gamma)}{\partial \gamma} = K_i \delta_i. \tag{18}$$

The minimum acceptable value of δ_i for recognizing the sign of $\partial I_s(\gamma)/\partial\gamma$ is considered and defined as $\delta_{i\min}$. To guarantee a stable operation of the IPMSM, the maximum changing range of the current vector angle in adjacent sampling periods is defined as $\Delta\gamma_{\max}$. In terms of (18), the parameter K_i is supposed to satisfy the following inequality:

$$|K_i| \le \frac{|\Delta\gamma_{\max}|}{T_s\,|\delta_{i\min}|}, \tag{19}$$

where T_s is the sampling period of the algorithm. It can be observed from (17) that the increase of K_i will reduce the convergence time of the algorithm. However, K_i should also obey the inequality in (19), and an excessively large value of K_i may result in undesirable fluctuations of $d\gamma/dt$ and currents, which may further lead to non-convergence of the algorithm and the instability of machine operation.

Without considering the sensor noise and interferences in a practical application, the algorithm may converge to an ideal stable operating point corresponding to the optimal current angle as shown in Fig. 6.

As seen in Fig. 6, the values of $I_s[k]$ and $I_s[k-1]$ may become the same and $\delta_i[k]$ reaches 0 near the optimal point γ_M. Hence, an error of $\pm A/2$ may exist after the convergence of the algorithm, which needs to be compensated.

C. Control System Configuration and Implementation

A block diagram of speed-servo IPMSM control system with the proposed step-signal-injection-based robust MTPA strategy is shown in Fig. 7. Therein, an outer-loop PI controller is employed to track the speed reference of the control system

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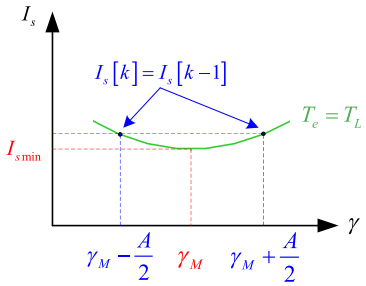


Fig. 6. Ideal stable operating point of the optimal current angle tracking.

and provide the current magnitude reference for the disturbance detection unit (DDU). After selecting the appropriate operating mode of the MTPA control, the reference values of i_q and i_d are calculated using (2). Two fast PI controllers are used internally in current loops to regulate the q- and d-axes currents of the IPMSM, which is fed by the three-phase voltage source inverter (VSI) controlled using space vector pulse-width modulation (SVPWM).

This paper assumes that disturbances in the IPMSM control system mainly consist of changes in load torque and speed reference, as well as variations in machine parameters caused by operating temperature changes or magnetic nonlinearities. However, not all of these disturbances affect the optimal current vector angle γ. As observed in Fig. 4, if a deviation of the current angle occurs from its optimal value, the current magnitude will also change. According to this feature, a 4-step disturbance detection strategy is proposed for the DDU.

Step 1: When the IPMSM is operating in steady-state under MTPA control, it records the last value of the current magnitude command that was produced by the optimal angle updating algorithm. If there is a change in operating conditions, the difference in current magnitude between the present state and the past optimal value is detected in real time and denoted by I_{dif}. If I_{dif} exceeds a pre-defined current threshold I_{th}, the local disturbance counter is incremented by 1. Otherwise, if I_{dif} is less than the threshold, the local disturbance counter will be reset to 0.

Step 2: When the number in the counter reaches a certain level (which is set to avoid potential interferences in the system), the disturbance flag f_D is set to 1, and the operation is switched into updating mode, where a positive step signal is first injected into the current angle according to Fig. 5.

Step 3: In the next sampling period after the first injection, if a positive δ_i is observed, the sign of the step appears to be incorrect according to Fig. 4, which indicates that a step in opposite direction should be injected instead.

Step 4: After the correct step signal is chosen, the proposed algorithm starts to continuously inject step signals (see Fig. 5) with the time interval of T_s. When the current vector angle finally converges within a small error tolerance, the obtained angle can be regarded as the optimal. At this time, the disturbance flag f_D will be reset to 0, and the controller is switched to steady-state MTPA operation mode.

It is noted that the sampling period T_s and the magnitude of the step A need to be chosen adequate for recognizing the sign of δ_i, which means that T_s should be longer than the response time of the current PI controller. Moreover, the threshold I_{th} should include sufficient margin to avoid a potential non-convergence of the algorithm during the first injection. Since the current vector angle γ is not influenced by rotor speed, acceleration and deceleration of the machine caused by speed reference changes are not designed to trigger the local disturbance counter in the DDU.

The advantage of the proposed algorithm is that it can be implemented without knowledge of the IPMSM electrical parameters and loading condition, while achieving high efficiency MTPA operation. Theoretically, only the phase currents magnitude and the speed will be slightly affected by the step-signal injection, and the operation of the machine will remain stable during updating the current angle to its optimal value. Moreover, complex computational process and digital filters are not required, which makes this approach suitable for implementation on low-cost microcontrollers.

IV. Experimental Verifications

A. Experimental Results on IPMSM

In this section, the proposed MTPA strategy is validated experimentally using IPMSM with the parameters summarized in Appendix A. The experimental setup consists of two coupled machines. The inverters of both IPMSMs are constituted by insulted gate bipolar transistors (IGBTs) with corresponding configurable controls. The first machine is used as the subject motor, and the second machine is operated in a constant-torque mode to generate the required mechanical load torque. The control unit of the subject IPMSM is realized using digital signal processor (DSP) TMS320F28335. Due to limitations of the experimental hardware and the safety considerations, the dc-link voltage of the inverters is regulated at 250V DC. The experimental data and waveforms are collected using a PC-based data-acquisition software of the DSP.

In the experiments with IPMSM, some related parameters of the proposed algorithm (see Fig. 5 and (19)) are designed as follows. The magnitude of the step signal γ_h is selected as 3 deg. To guarantee a stable operation of the machine, the maximum changing range of the current vector angle $\Delta\gamma_{\max}$ in two consecutive sampling periods is determined by 5 deg, and the sampling period T_s and minimum acceptable value of δ_i for recognizing the current magnitude gradient are selected as 0.1 s and 0.3 A, respectively.

In the following study, the IPMSM initially operates at 1200 r/min with the load torque of 5 Nm. To illustrate the effectiveness of the proposed optimal current angle tracking method, the initial current angle γ (see Fig. 3) is set to 120 deg (which is not optimal for the considered operating condition). Then, the proposed algorithm is enabled to online search for the optimal current under the MTPA operation. The experimental results with the current angle, current magnitude, q- and d-axes currents, and rotor speed are shown in Fig. 8.

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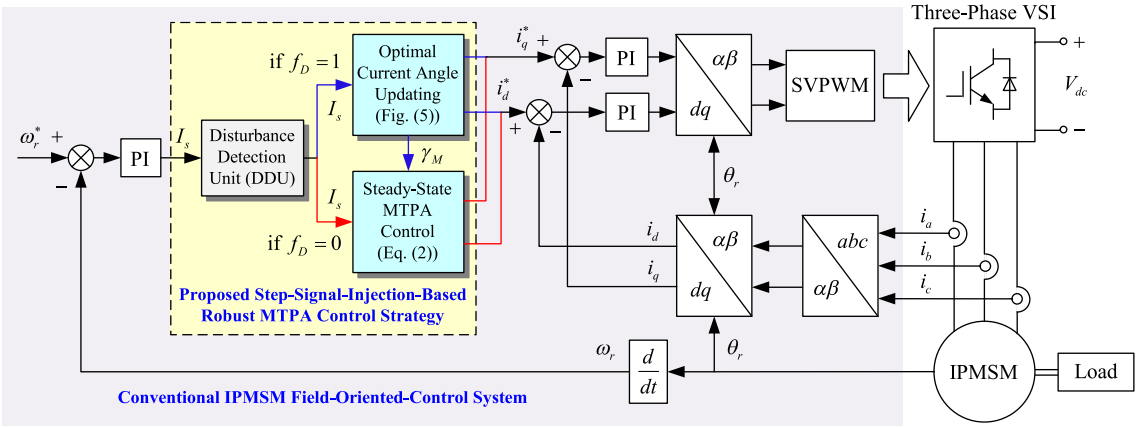


Fig. 7. Block diagram of the IPMSM control system with the proposed step-signal-injection-based robust MTPA control strategy.

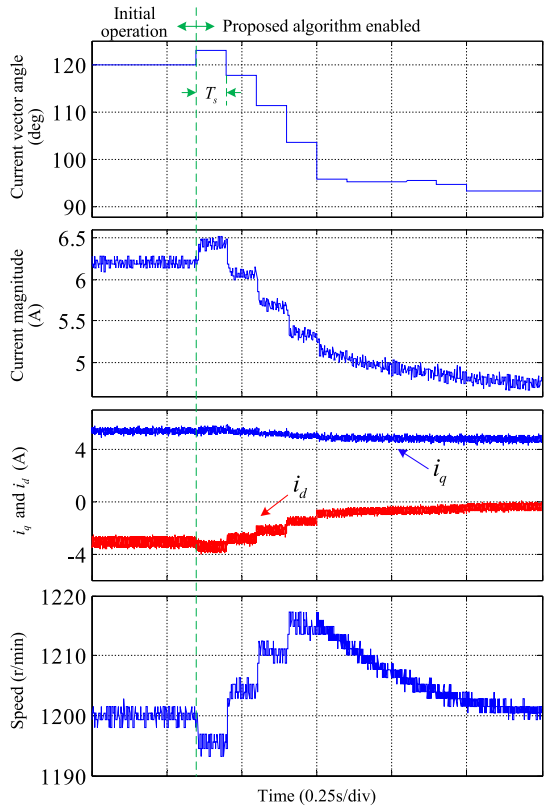


Fig. 8. Experimental results demonstrating optimal current vector angle tracking for the IPMSM loaded at 5 Nm.

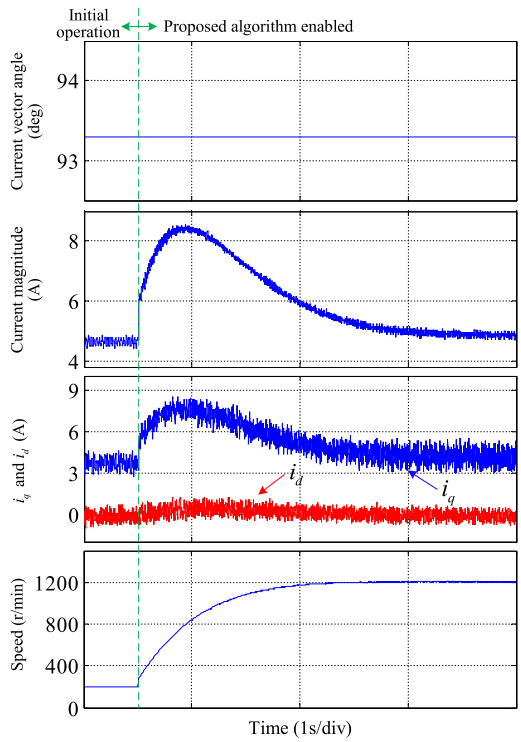


Fig. 9. Experimental results demonstrating response to step change in speed reference from 200 rpm to 1200 rpm for the IPMSM loaded at 5 Nm.

As it can be observed in Fig. 8, before the algorithm starts, the current vector angle is at 120 deg and the current magnitude is approximately 6.2 A. After the algorithm is enabled, the first injection step is taken to increase the angle to 123 deg. This increase in angle results in increase of the current magnitude to approximately 6.4 A, which is recognized by the algorithm as the step in opposite direction. After that, the algorithm takes several steps and subsequently reduces to the optimal current angle to 93.3 deg (with the relative error of approximately 0.5%), leading to reduction of current magnitude to 4.5 A. As it can be observed in Fig. 8, it takes the proposed algorithm approximately 9 sampling periods (0.9 s) to attain the optimal MTPA operation of the IPMSM. During the step signal injection process, the motor speed fluctuates slightly (1.3%) and retunes to 1200 rpm after 1 s. These experimental results verify the effectiveness of the proposed robust MTPA algorithm.

In the next study, a step change in speed reference from 200 r/min to 1200 r/min is performed on the IPMSM motor loaded with the torque of 5 Nm. As shown in Fig. 9, the machine operates with the optimal current vector angle of 93.3 deg that has been derived in the first study (see Fig. 8). Fig. 9 demonstrates the measured responses of the currents and speed.

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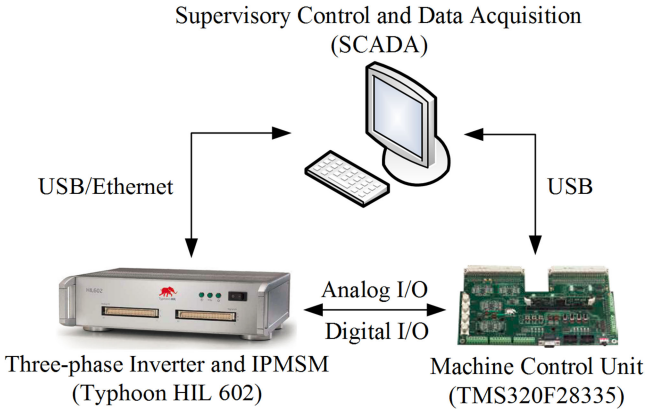


Fig. 10. Diagram of the HIL-based experimental setup.

As seen in Fig. 9, the q- and d-axes currents of the IPMSM rises during acceleration, and the rotor speed reaches 1200 r/min after approximately 1.5 s with the speed closed-loop PI regulation. It can be observed that the optimal current vector angle remains at 93.3 deg with the increase of the rotor speed. This is due to the fact that the acceleration of the machine caused by speed reference change does not affect the optimal current angle and it is not designed to trigger the local disturbance counter. When the rotor speed finally reaches the reference of 1200 rpm, the current magnitude recovers to the previous level. Hence, it is detected by DDU that the current magnitude difference I_{dif} is lower than the corresponding threshold in *Step 1* of the disturbance detection process, and the updating of the current vector angle is not required. Hence, it is verified from Fig. 9 that a robustness MTPA operation of the IPMSM under speed reference change can be ensured by using the proposed algorithm.

B. HIL-Based Experimental Validations

The IPMSM with the parameters summarized in Appendix A has fixed parameters, and is not with obvious saliency effect (low $L_{q,d}$ values, load torques, and current levels), which means that some of the merits of the proposed algorithm (such as robustness under machine parameter variations and load torque changes) may not be appropriately demonstrated using this machine. Therefore, an HIL-based experimental setup shown in Fig. 10 is established to demonstrate the effectiveness and robustness of the proposed algorithm under extrapolated operation conditions and changes in parameters (which would be otherwise difficult to implement experimentally without modifying or damaging the IPMSM). Therefore, the IPMSM is emulated on a Typhoon HIL 602 emulator, where it is more convenient to change the parameters of the machine and emulate multiple system perturbations [31], [32]. The proposed control is implemented on the actual TMS320F28335 DSP to verify the proposed robust MTPA strategy shown in Fig. 7. The parameters of the IPMSM and various controllers are summarized in Appendix B. Data and waveforms are collected by an oscilloscope and the PC-based SCADA system.

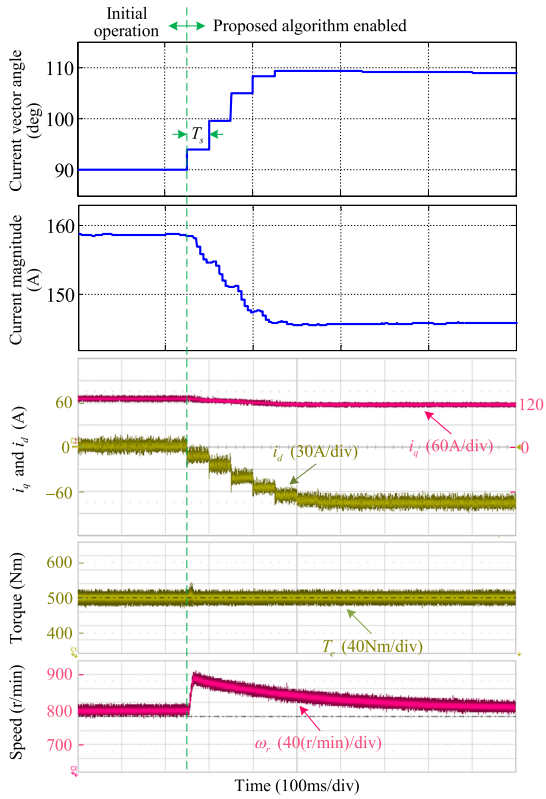


Fig. 11. HIL-based experimental results for optimal current angle tracking.

In the HIL-based experiments, the sampling period for the proposed algorithm is selected as 50 ms. Similar to the experiments with IPMSM, the magnitude of the step signal, the maximum changing range of the current vector angle $\Delta\gamma_{\max}$, and the minimum acceptable value of δ_i are determined by 3 deg, 10 deg, and 3A, respectively.

In the following study, the IPMSM is assumed to operate initially with zero current in d-axis under the load torque of 500 Nm and speed of 800 r/min. Then, the proposed algorithm is enabled to online search for the optimal current under the MTPA operation. The recorded responses in the current angle, current magnitude, q- and d-axes currents, electromagnetic torque, and rotor speed are shown in Fig. 11.

As it can be observed in Fig. 11, before the algorithm is enabled, the current vector angle is at 90 deg while the d-axis current remains at 0 A. Under this condition, only the field torque contributes to the total electromagnetic torque of the IPMSM according to (1). After the proposed algorithm is enabled, the optimal current angle is obtained after several sampling periods and reaches approximately 109.6 deg (with the relative error of 0.9%), and the current magnitude decreases from 158 A to 146 A, respectively. The d-axis current goes to -43 A, which indicates that the reluctance torque also contributes to the production of the overall electromagnetic torque. During the step signal injection process, the motor speed fluctuates slightly and recovers after 1s, while the electromagnetic torque remains stable at the desired load torque 500 Nm. Overall, as seen in Fig. 11, it takes the proposed algorithm approximately 5 sampling periods (0.25 s) to attain MTPA operation of the considered

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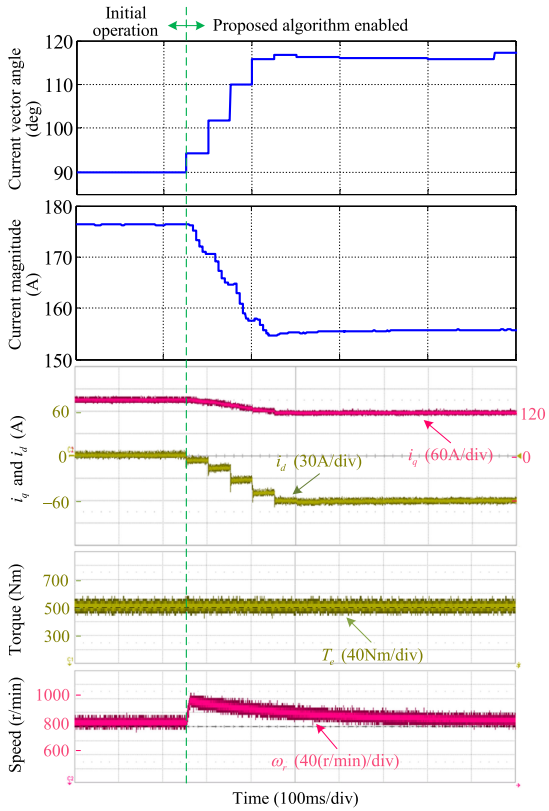


Fig. 12. Experimental results for online optimal current angle tracking with IPMSM stator resistance and q- and d-axes inductances increased by 20%, and flux linkage of permanent magnet decreased by 20% (see Appendix B).

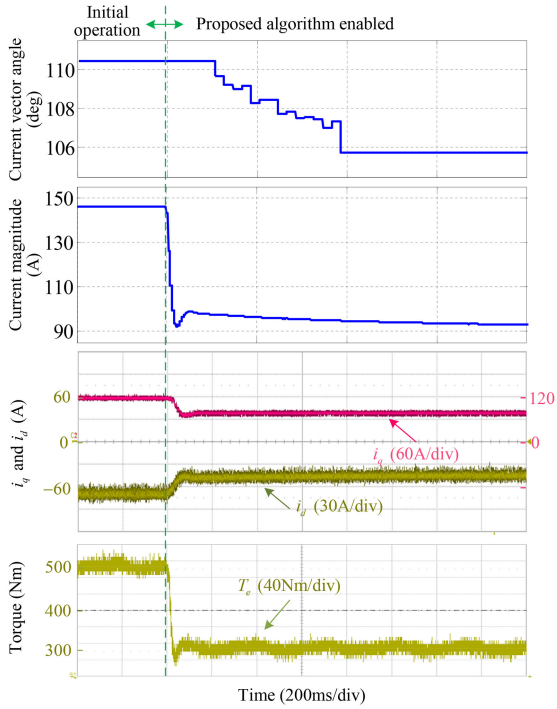


Fig. 13. Experimental results for online optimal current angle tracking under load change.

IPMSM. This case also verifies the effectiveness of the proposed step-signal-injection-based robust MTPA algorithm for a larger machine with different parameters.

To further examine the effectiveness of the proposed algorithm under significant perturbations in machine parameters, it is assumed that (e.g., due to an elevated temperature) the stator resistance and q- and d-axes inductances of the IPMSM have been increased by 20%, and the flux linkage of the permanent magnet has been decreased by 20% (see Appendix B). The variations in inductances caused by magnetic saturations and cross-coupling effects can be emulated similarly. The results obtained on the HIL platform with the actual DSP controller are illustrated in Fig. 12. As it can be seen in Fig 12, after 5 sampling periods (0.25 s), the current vector angle is regulated to its optimal value at approximately 116.3 deg, while the current magnitude has been reduced from 176 A to 155 A. The relative error of the optimal current angle is calculated approximately as 0.8%. As it can be observed, despite slight fluctuations in the speed during signal injection, the proposed algorithm accurately achieves the optimal MTPA operation.

Next, the load torque change is also considered in the HIL-based experiment. In this scenario, the IPMSM operates at 800 r/min, while the load torque is decreased from 500 Nm to 300 Nm. The resulting current vector angle, current magnitude, q- and d-axes currents, and torque are shown in Fig. 13.

As it can be observed in Fig. 13, when the torque is changed from 500 Nm to 300 Nm, the current magnitude is also decreased to achieve a power balance. The DDU described in Fig. 7 detects the difference I_{dif} in *Step 1*. When the flag f_D is set to 1 in *Step 2*, the control system is switched into optimal current angle updating mode, where the step signal is injected into the current angle. As it can be seen in Fig. 13, after 14 sampling periods (0.7 s), the optimal current vector angle is updated from 110.2 deg to 105.7 deg (with the relative error of approximately 0.7%), and the current magnitude decreases to its new optimal value at 92 A. These results also verify the robustness of the proposed robust MTPA operation strategy under load change condition.

V. Conclusions

To mitigate the drifting of the MTPA operation of the IPMSMs caused by system perturbations and variations in parameters, a step-signal-injection-based robust MTPA control method is proposed employing a step perturbation of the stator current vector angle. The optimal current angle is obtained by observing the response in the current magnitude gradient and subsequently minimizing the current using a PI controller, and therefore the proposed methodology does not require electrical parameters of the machine. The proposed algorithm is demonstrated on a speed-servo control system, where an optimal current angle updating mode and a steady-state MTPA operation mode can be selected according to the presence of disturbance in the system.

The proposed methodology is experimentally verified on a practical IPMSM and a HIL-based machine with the actual controller implemented in DSP to validate the effectiveness and robustness of the proposed algorithm. Based on the presented results, it can be concluded that an optimal MTPA operation

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of the IPMSM can be achieved under variations in machine parameters, load torque changes, and speed changes, without using electrical parameters of IPMSM. Future works may be considered with simplifying the algorithm and including field-weakening operations of the IPMSM.

Appendix A
Parameters of Experimental IPMSM and Its Controller

Parameters	Values
Rated power	2.3 kW
Rated current	9.5 A
Rated voltage	220 V
Rated speed	1500 r/min
Pole pairs P	4
Stator resistance R	0.628 Ω
q-axis inductance L_q	4.117 mH
d-axis inductance L_d	1.781 mH
Flux linkage λ_f	0.229 Wb
DC-link voltage	250 V
Parameters K_p and K_i of the proposed algorithm	5, ±50
Amplitude of the step signal	3 deg
Current threshold I_{th}	0.8 A
Parameters K_{p1} and K_{i1} for speed control	0.05, 0.1
Parameters K_{p2} and K_{i2} for current control	8, 15
Sampling period of the proposed algorithm	0.1 s
Sampling frequency of the control system	10 kHz

Appendix B
Parameters for HIL-Based Emulated IPMSM and Its Controller

Parameters	Values
Rated power	100kW
Maximum current	200A
DC-link voltage	1500V
Rated speed	1500r/min
Pole pairs P	4
Stator resistance $R_{1,2}$	0.05Ω, 0.06Ω
q-axis inductance $L_{q1,2}$	4.1mH, 4.9mH
d-axis inductance $L_{d1,2}$	1.6mH, 2.0mH
Flux linkage $\lambda_{f1,2}$	0.65Wb, 0.52Wb
Parameters K_p and K_i of the proposed algorithm	5, ±60
Amplitude of the step signal	3deg
Current threshold I_{th}	5A
Parameters K_{p1} and K_{i1} for speed control	0.15, 0.025
Parameters K_{p2} and K_{i2} for current control	12, 30
Sampling period of the proposed algorithm	0.05s
Sampling frequency of the control system	10kHz

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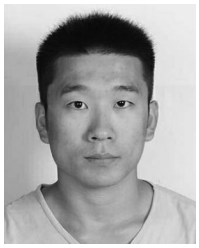
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